

Orbit Determination Report
2005 EC224

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Abstract

Orbit Determination of Near Earth Asteroid 2005 EC224

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Determination of the orbit of a Near Earth Asteroid is not only important to humanity's future, but gaining knowledge of an asteroid's location is also critical for mission planning and satellite communications. In our research, we calculated the six orbital elements of Near Earth Asteroid 2005 EC224 by imaging it over a period of five weeks and analyzing our data, eventually comparing our values to those taken from JPL Horizons. After successfully seeing the asteroid during three observing sessions, we wrote Python scripts to perform astrometry, photometry, and the Method of Gauss, in order to calculate the position and velocity vectors of the asteroid at the location and time of the sighting. The calculated orbital elements were as follows: the semi-major axis was 2.1331, the eccentricity was 0.3239, the inclination was 2.787, the Argument of the Perihelion 74.485, the Longitude of the Ascending Node 163.7502, and the Mean Anomaly was 22.1. The percent error of our calculated orbital elements was, on average, 6.7940% from those of JPL Horizons. Using our images, we calculated the vectors and orbital elements, then allowing us to calculate the position of the asteroid relative to the Earth at any given time.

Introduction

Near Earth Asteroids are a subset of asteroids, and are typically found in the main asteroid belt between Mars and Jupiter. These asteroids are defined by an orbital path that causes them to pass closer to Earth's orbit than any other planet does. In our research, we took images of a Near Earth Asteroid, 2005 EC224, over a period of 5 weeks. We then analyzed this data, and used the Method of Gauss to determine its orbital elements. We hypothesized that the calculated orbital elements will adequately align with the theoretical ones, as obtained from JPL Horizons, and displayed later in our Analysis Section, Table 3. Since our goal is to calculate orbital elements accurately, our calculated elements should ideally include the theoretical JPL values in their error ranges, which will be our measurement of success. However, even if we do not succeed, simply getting elements near the theoretical ones will still be considered an accomplishment for our team.

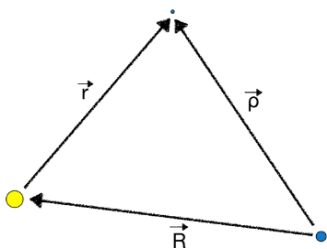
Calculating the orbital elements of our asteroid was a multistep process. We started with raw science images of our asteroid, which were then processed using astrometry, in order to get the asteroid's position in the sky, in terms of right ascension (RA) and declination (Dec). We also used aperture photometry to calculate the approximate brightness of the asteroid. Using the three sets of calculated RA and Dec values, we used the Method of Gauss to calculate our position and velocity vectors. Following this step, we used the two vectors to calculate our asteroid's orbital elements.

Method of Gauss

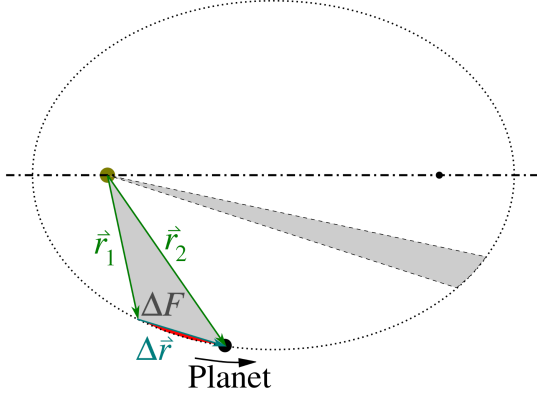
The Method of Gauss relies on the basic principles of vector addition and subtraction. Starting off with the known Sun-Earth vectors ($\mathbf{R}$) and computing the Earth-Asteroid vectors ($\mathbf{p}$) with the help of our data at the time of the observation, we can calculate the Asteroid-Sun vectors ($\mathbf{r}$). The $\mathbf{r}$ vector can be worked out by subtracting the $\mathbf{R}$ vector from the $\mathbf{p}$ vector:

$$\vec{r} = \vec{p} - \vec{R}$$

To calculate the $\mathbf{p}$ vector, we need the earth asteroid range ($\mathbf{p}$) and its unit vector ($\hat{\mathbf{p}}$).



$$\vec{r} = \rho \hat{\rho} - \vec{R}$$



Next, with the help of Kepler's Second Law, we get a first approximation for the position and velocity $\vec{r}$ vectors. In Kepler's second law, we estimate the area swept out between the first and second, second and third, and first and third observations (B1, B3, and B, respectively) as triangles (A1, A3, and A, respectively) and compute them by using the cross product between the corresponding $\vec{r}$ vectors:

$$A = \frac{1}{2} |\vec{r}_1 \times \vec{r}_3|$$

$$A_1 = \frac{1}{2} |\vec{r}_1 \times \vec{r}_2|$$

$$A_3 = \frac{1}{2} |\vec{r}_2 \times \vec{r}_3|$$

Then, we use the fact that $\vec{r}_2 = c_1 \vec{r}_1 + c_3 \vec{r}_3$ (92) to get

$$A_1 = \frac{1}{2} |\vec{r}_1 \times (c_1 \vec{r}_1 + c_3 \vec{r}_3)| = \frac{1}{2} c_3 |\vec{r}_1 \times \vec{r}_3| = c_3 A$$

$$A_3 = \frac{1}{2} |(c_1 \vec{r}_1 + c_3 \vec{r}_3) \times \vec{r}_3| = \frac{1}{2} c_1 |\vec{r}_1 \times \vec{r}_3| = c_1 A$$

Therefore,

$$c_1 = \frac{A_3}{A} \simeq \frac{B_3}{B} = \frac{t_3 - t_2}{t_3 - t_1} = \frac{\tau_3}{\tau}$$

$$c_3 = \frac{A_1}{A} \simeq \frac{B_1}{B} = \frac{t_2 - t_1}{t_3 - t_1} = \frac{-\tau_1}{\tau}$$

Now, we simply use (92) to get an estimation for $\vec{r}_2$. To calculate $\dot{\vec{r}}_2$,

$$\dot{\vec{r}}_2 \simeq \frac{\Delta \vec{r}}{t} \simeq \frac{\vec{r}_3 - \vec{r}_1}{\tau}$$

After getting a first guess for the position and velocity vectors, we used a fourth order truncated Taylor series expansion of $\vec{r}$ about central value to obtain our position and velocity vectors:

$$f = 1 - \frac{1}{2} u \tau^2 + \frac{1}{2} u z \tau^3 + \frac{1}{24} (3uq - 15uz^2 + u^2) \tau^4 + \dots$$

$$g = \tau - \frac{1}{6} u \tau^3 + \frac{1}{4} u z \tau^4 + \dots$$

$$u = \frac{\mu}{r_2^3}; \quad z = \frac{\vec{r}_2 \cdot \dot{\vec{r}}_2}{r_2^2}; \quad q = \frac{\dot{\vec{r}}_2 \cdot \dot{\vec{r}}_2}{r_2^2} - u$$

We used these equations to solve for f and g , then solved for the c and d values, as well as the ρ values. Then, we calculated the position and velocity vectors. We looped through this process until we

achieved consecutive position vectors that were within a set tolerance of each other. Once this process converged, we used the position and velocity vectors to calculate the orbital elements.

Orbital Elements

The six orbital elements we calculated were a , the semi-major axis of 2005 EC224's orbit, e , the eccentricity of the orbit, I , the inclination of the object's orbital plane with respect to the ecliptic, Ω , the longitude of the ascending node, ω , the argument of perihelion, and M_0 , the mean anomaly at the epoch time, which we set to be 6:00 UTC on July 23rd, 2023. We take the position and velocity vectors of our asteroid at the middle observation and use these values to calculate all orbital elements. Each of these elements were calculated using several equations, all of which are explained and listed in Appendix C.

Once we each calculated our orbital elements, we all ran a Monte Carlo simulation to estimate uncertainty values on each of the orbital elements. This simulation involved, essentially, using slightly altered RA and Dec values and repeatedly recalculating the orbital elements. For each of the RA and Dec values throughout all observations, we generated around 1 million random different values in the normal distribution with the RA or Dec value as the mean, and the value's error as the standard deviation. For each one of these 1 million elements, we calculated the orbital elements, recording them down in a list, with a different list for each element. Then, we took each of the lists and calculated the standard deviation, which represented our uncertainties for each of the calculated orbital elements. The equation used by our Python code in calculating the standard deviation is as follows:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

where σ is the standard deviation, x_i represents each value in the list, μ represents the mean of the dataset, and N is the size of the dataset.

Equipment and Procedure:

To obtain at least three clear sightings to use the Method of Gauss for the eventual orbit determination of EC224, our team used the Tortugas Mountain Observatory's 1967 Boller and Chiven 24" (60.96 cm) reflector telescope, with a focal ratio of $f/5.6$. The telescope has a 6" two-element refractor guide scope mounted at its side, while the instrumentation is mounted at the front of the telescope- at the Newtonian focus. The telescope is an equatorial mount with counterweights for the Right Ascension in the base and those for the Declination attached to the back end of the tube. The '67 B&C telescope has a QHY 600 Complementary Metal-Oxide-Semiconductor (CMOS) camera that utilizes a Sony IMX 455 sensor, with a 9,576 x 6,388 array of 3.76 micron pixels. While the QHY software allows for 1x1 and 2x2 binning, due to the small pixels, mainly the 2x2 binning is used. The AstroTech Coma Corrector is in front of the camera, pushing focus back (making the $f/5$ an essentially $f/5.6$ telescope). The camera chip is cooled to a temperature of -12°C thermoelectrically.

For data collection, Maxim DL is primarily used. Not only does it pilot the camera, but it also operates the filter-wheels, focusers, and autoguider. The TMO B&C uses MaximDL 6.20, ACP/ACP Scheduler (8.2.1)- mainly used for dome control and to check pointing-, SiTechExe (0.94N) plus the SiTech Servo II Driver (9.7)- primarily for the automated tracking of the asteroid by the telescope- alongside FocusMax v4- used for occasionally automated and mostly manual focusing.

After successfully following pre-established TMO procedures for connecting to the telescope, camera, and dome, followed by the steps for pointing checking and manually focusing, the telescope can be slewed to the coordinates of the object of interest- EC 224. Starting off by setting a path

directory to save the images and ensuring the binning is 2x2, the Lights and Darks could finally be taken. Since we weren't scaling our darks, our team had no need to proceed with taking biases. For each set of data- the Lights- a series of 5 1-minute exposures were taken, initially with the 'V'; and later on with 'Open' filter- due to how faint the asteroid is- over an, on average, 2 hour period. The same procedure was followed for darks, with the only change being the change in filter. The darks are still classified as 'Lights' under image type, due to the fact that the CMOS camera does not have a shutter, and therefore the dark filter needs to be used. The 1 minute exposure darks are also taken in sets of 5, and are all taken at the same °C temperature of the camera cooler. For our team, we decided to take a median of 4 sets of lights and 1 set of darks every observing shift- that were every 4 days-, with the lights being the first and last sets of data we took during our observing shift- as seeing conditions permitted throughout the shift. Our team was, on occasion, unable to take adequate data due to cloudy atmospheric conditions, but thankfully had clear skies for a majority of our observing shifts.

Data

Date	Time (UTC)	RA (RMS Uncert.) (h:m:s)	Declination (RMS Uncert.) (h:m:s)	Apparent Mag	Inst. Mag	SNR
06-26-2023	05:54:36.74	19:02:34.16 ± 0.11	-11:35:08.540 ± 0.003	16.9	-8.66 ± 0.10	10.682
06-26-2023	6:45:33.30	19:02:34.4 ± 0.5	-11:35:30.7 ± 0.4	17.0	-8.50 ± 0.14	7.607
07-01-2023	07:39:20.81	19:03:33.8 ± 0.4	-12:03:04.9 ± 0.2	16.8	-10.10 ± 0.06	19.239
07-01-2023	08:54:52.42	19:03:33.7 ± 0.7	-12:03:29.1 ± 0.5	16.8	-9.55 ± 0.09	12.448
07-17-2023	05:35:30.16	19:05:30.0 ± 0.2	-14:03:18.17 ± 0.2	17.3	-9.70 ± 0.04	25.460
07-17-2023	05:50:41.67	19:05:29.93 ± 0.06	-14:03:23.35 ± 0.03	17.3	-9.57 ± 0.04	19.239

For determining the orbit of an asteroid through the Method of Gauss, a minimum of 3 observations are needed. The following information is recorded and calculated for those 3 observations: the date and time (in UTC), the Right Ascension and Declination, the apparent magnitude and the asteroid Signal-to-Noise Ratio. While the date is simply noted, the time is documented by adding half the time of the exposure to the time when the image had started to be taken- the time recorded is that of the exact midpoint of the image where the asteroid was sighted. The RA and Declination are recorded after doing astrometry on the images and a python code was used to calculate the Root-Mean-Square deviation of the RA and Dec in arcseconds. The apparent, instrumental magnitude and Signal-to-Noise Ratio were calculated through photometry- plotting a graph of apparent magnitude over a log10 of the signal and calculating the dark current value.

Analysis

After getting our raw science images, we used the flat images taken by Dr. R, as well as the darks we took during our observing session, to reduce our images. We used SAOImage DS9 to build a master dark image for our flats, then subtract this from our flat images. The flat images are used to correct for optical aberrations present in our images, such as dust spots, or vignetting. Then, we used our darks from observing to build a master dark, then subtracted this from our raw images. Subtracting the dark current from our images removes signal due to heat that is captured by the camera during imaging. The result was a set of much clearer science images, ones with less hot pixels and vignetting.

Upon reducing our data, we performed astrometry to get the RA and Declination values of our asteroid. We uploaded the reduced images to nova.astrometry.net, and obtained a processed image with RA and Dec values labeled on the image, when we opened it in AstroImageJ. Navigating to and finding our asteroid, we recorded the approximate RA and Dec values displayed on the screen. We also used the corr.fits file provided by nova in order to calculate the RMS uncertainty values for each of our RA and Dec statistics. The obtained RA and Dec values from our observing sessions, as well as their uncertainties, are shown above in Table 1, columns three to four.

Following astrometry, we performed aperture photometry on our images, in order to estimate the asteroid's brightness, to be used in our MPC submission. This process was a rather time consuming one, as we had to open a bright star catalog using SAOImage DS9, and find 7-10 reference stars. We recorded the brightness of the star, as well as its respective (x, y) position in our reduced image. Then, for each of these reference stars, we used a Python script to calculate the log (base 10) of its signal. We then used Google Sheets to build a linear model using this data, relating the log of signal to the apparent magnitude of the bright stars, and obtained a linear regression equation relating log signal to apparent magnitude. Finally, we applied this model to our asteroid, after calculating the log of its signal, using similar code. The result was the calculated apparent magnitude of our asteroid.

Image	Date	Time (UTC)	Slope	Y-intercept	Appt. Mag	Inst. Mag	Dark Current	SNR
Obs1, Set1, Img0	06-26-2023	05:54:36.74	-2.61 ± 0.06	25.9 ± 0.3	16.9	-8.66 ± 0.10	6.795	10.682
Obs1, Set1, Img5	06-26-2023	6:45:33.30	1.0 ± 0.1	25.4 ± 1.0	17.0	-8.50 ± 0.14	6.795	7.607
Obs3, Set1, Img 1	07-01-2023	07:39:20.81	-2.60 ± 0.06	27.3 ± 0.3	16.8	-10.10 ± 0.06	6.717	19.239
Obs3, Set5, Img4	07-01-2023	08:54:52.42	0.96 ± 0.07	26.0 ± 0.7	16.8	-9.55 ± 0.09	6.717	12.448
Obs7, Set6, Img3	07-17-2023	05:35:30.16	1.02 ± 0.06	27.3 ± 0.7	17.3	-9.70 ± 0.04	6.717	25.460

Obs7, Set8, Img1	07-17-2023	05:50:41.67	-2.52 ± 0.06	26.9 ± 0.03	17.3	-9.57 $\pm$ 0.04	6.717	19.239
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Table 2: A table summarizing our photometry data, including the slope and intercept of the linear regression fit.

In the above table, for a total of six data points for the three sightings of 2005 EC224, aperture photometry was performed on them. The following was recorded: file name of the image, the date and time (in UTC), the slope of the linear model relating the log10 of the signal to the apparent magnitude of the asteroid, the Y-intercept of said model, and the calculated apparent and instrumental magnitudes of our asteroid, alongside the dark current of the image and the Signal-to-Noise Ratio of the asteroid.

Each team member also calculated their own orbital elements, using the Method of Gauss, calculating the elements themselves, and also running Monte Carlo to determine the uncertainties of each value. Below is a table displaying the results from each team member. The value in parentheses following each team member's name represents the number of Monte Carlo iterations performed by the specific person.

Orbital Element	Rajat (1,000,000)		Isbaah (500,000)		Vivian (1,000,000)		JPL Prediction
	Value	Uncert	Value	Uncert	Value	Uncert	
a (au)	1.90148	0.06985	2.1331	0.0013	1.91061	0.04831	2.057670
e	0.35313	0.01494	0.3239	0.0003	0.355645	0.010123	0.386466
i ($^{\circ}$)	2.34780	0.13387	2.787	0.004	2.36539	0.09190	2.624589
ω ($^{\circ}$)	73.349	3.153	74.485	0.003	73.8036	2.1364	78.87999
Ω ($^{\circ}$)	172.976	2.023	163.7502	0.0014	172.6607	1.3589	169.35866
$M(t_0)$ ($^{\circ}$)	23.989	1.591	22.1	0.3	23.7146	1.0677	20.88101

Table 3: The different orbital elements and uncertainty calculated by each team member, as well as the expected values.

In the above table, the uncertainties were obtained using the Monte Carlo method. The first column represents the different orbital elements we're calculating, with the units. The second and third columns are the values and uncertainties for each of these elements, as Rajat calculated. The fourth and fifth columns are Isbaah's calculations, and the sixth and seventh columns are Vivian's calculations. The final column is the predicted values from JPL, which we use as the expected orbital elements.

In running Monte Carlo, we calculated the orbital elements many, many times. The elements produced formed a Normal distribution, as shown in the below histograms of our data.

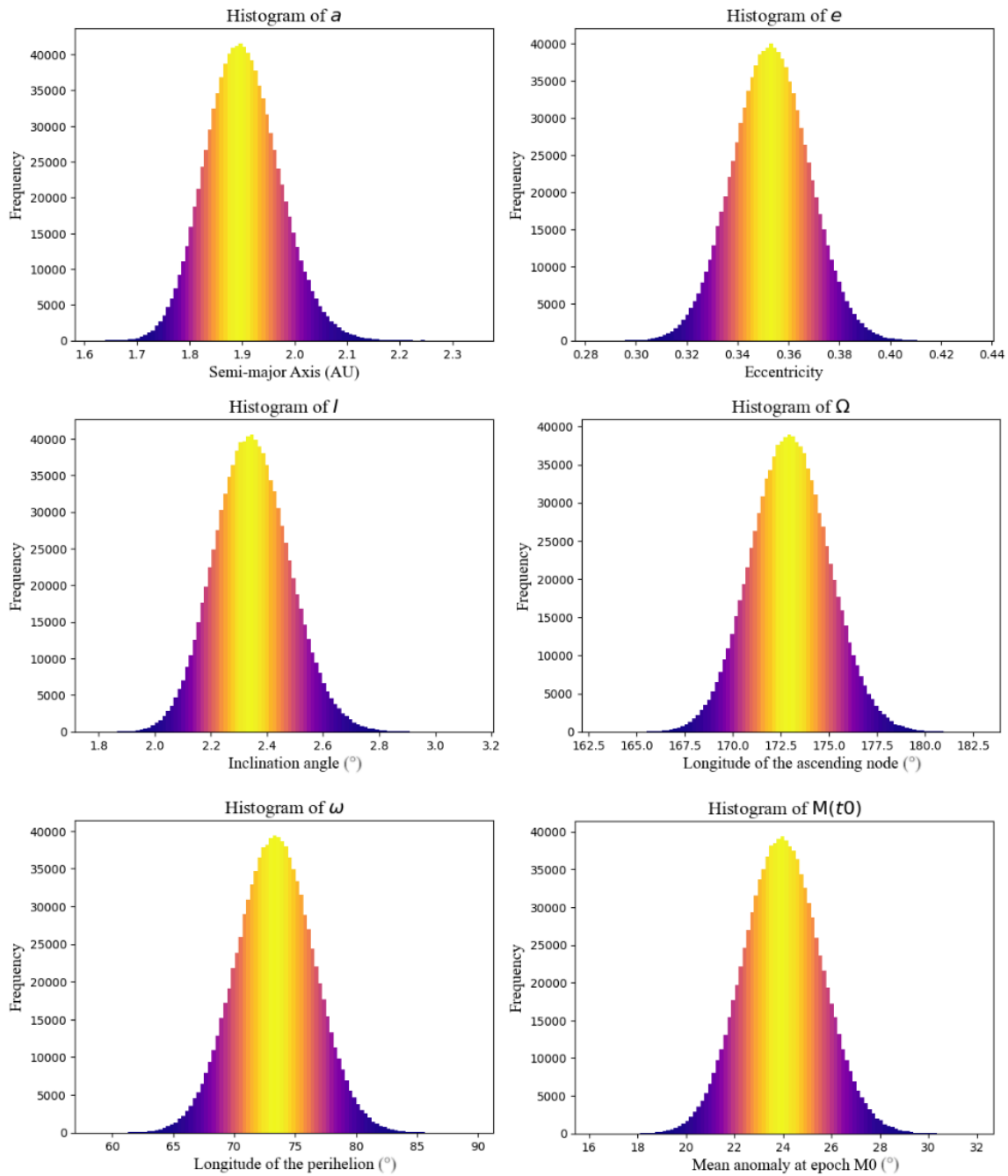


Figure 1: Histograms of orbital elements with Monte Carlo (1 million iterations)

When looking at our final data, we calculated the percent error of each team member's orbital elements, with respect to the JPL expected values. We then took the set of orbital elements that had the lowest average percent error, which ended up being Isbaah's, and we will use these values as our official ones. Below is a table of our percent error values, as well as the average percent errors:

Orbital Element	Rajat (1,000,000)		Isbaah (500,000)		Vivian (1,000,000)		JPL Prediction
	Value	% Error	Value	% Error	Value	% Error	
a (au)	1.90148	7.59062	2.1331	3.6658	1.91061	7.14692	2.057670
e	0.35313	8.62586	0.3239	16.1893	0.355645	7.9750871	0.386466
i (°)	2.3478	10.5460	2.787	6.188	2.36539	9.87579	2.624589
ω (°)	73.349	7.012	74.485	5.572	73.8036	6.4356	78.87999
Ω (°)	172.976	2.136	163.7502	3.3116	172.6607	1.9497	169.35866
$M(t_0)$ (°)	23.989	14.884	22.1	5.8	23.7146	13.5702	20.88101
Average % Error		8.4658		6.7940		7.8255	

Table 4: The orbital elements by each team member, and the corresponding percent errors, compared to the rightmost expected values.

In the above table, the first column represents the orbital element, and each subsequent pair of columns represents the values and percent errors calculated by each team member. The sixth and seventh columns contain the orbital elements and percent error that we officially use as our own, and the last column includes the expected JPL values.

Now, we can incorporate the uncertainty values of our data and create a range of values, in order to check if the theoretical orbit elements fall in this range:

Orbital Element	Value	Uncert	Range	JPL Predicted Values	Is the predicted value in range?
a (au)	2.1331	0.0013	(2.1318, 2.1344)	2.057670	No
e	0.3239	0.0003	(0.3236, 0.3242)	0.386466	No
i (°)	2.787	0.004	(2.783, 2.791)	2.624589	No
ω (°)	74.485	0.003	(74.482, 74.488)	78.87999	No
Ω (°)	163.7502	0.0014	(163.7488, 163.7516)	169.35866	No
$M(t_0)$ (°)	22.1	0.3	(21.8, 22.4)	20.88101	No

Table 5: Orbital elements and their respective ranges, based on uncertainty values.

In the above table, the first column is the orbital element, and the second and third are our official orbital elements. The range was calculated by taking the uncertainty and adding to and subtracting it from our orbital element value. The fifth column is the JPL predicted values, and the sixth column tracks if the values are in the range or not. Unfortunately, none of the ranges for the orbital elements included the JPL predicted values. Though this was disappointing, we are still glad that our data is relatively close to the JPL predicted values, and consider it a success of our overall project. A possible cause of this difference could be due to the fact that our three observations were

not equally spaced out. Our first two data points in MoG were fairly close together, only about 5 days apart. Between our second and third data point, however, we had around 14 days of spacing, meaning that it might be harder for our data to fit to an accurate elliptical orbit.

Conclusion

As mentioned earlier, calculating the orbital elements of asteroid 2005 EC224 was a multistep process. For our orbital elements, the average percent error from the JPL Horizons predicted values—our hypothesized values—was about 6.7940%. For the semi-major axis, our final value was 2.1331 au, and when compared with the predicted JPL value of 2.057670 au, gave an error of 3.67%. Our semi-major axis was not within the calculated uncertainty of 0.0013. For the eccentricity, our final value was 0.3239, and when compared with the predicted JPL value of 0.386466, gave us an error of 16.1893%. Our eccentricity was not within the calculated uncertainty of 0.0003. For the inclination, our final value was 2.787° , and when compared with the predicted JPL value of 2.624589° , gave us an error of 6.188%. It was not within the calculated uncertainty of 0.004. For the Argument of Perihelion, our final value was 74.485° , and when compared with the predicted JPL value of 78.87999° , gave us an error of 5.572%. It was not within the calculated uncertainty of 0.003. For the Longitude of the Ascending Node, our final value was 163.7502° , and when compared with the predicted JPL value of 169.35866° , gave us an error of 3.3116%. It was not within the calculated uncertainty of 0.0014. For the Mean Anomaly, our final value was 22.1° , and when compared with the predicted JPL value of 20.88101° , gave us an error of 5.8%. It was not within the calculated uncertainty of 0.3.

There were several sources of possible error in our experiment. For one, we were not able to space our observations evenly, with around 2 weeks between our second and third observations. As a result, the orbit that we determined is naturally biased toward the first two observations, which may not have been accurate due to the discrepancies in the orbit of EC224 (since asteroids do not orbit in perfect ellipses). This may have been improved if we were able to successfully run orbit improvement using additional data from Team 7. Unfortunately, we were not able to perform differential correction of our orbit due to issues with the Python code. However, much of the data from Team 7 had either incredibly faint sightings of the asteroid or were very close in time to one of our three sightings. Thus, it remains unclear whether running differential correction would have even improved our results. In addition, every piece of data that was used to compute the orbital elements has an associated uncertainty. The images taken from the telescope had a significant amount of noise, and the reduction process did not remove all of it. Therefore, centroiding, and thus astrometry and photometry may have been slightly off of the true value. With each calculation performed on the original data, uncertainty is also added. Thus, each of our orbital elements has uncertainty as well.

In the very near future, our calculated orbital elements will be used to determine how EC224's orbit will change over a large period of time with the arrival of the Southwest Research Institute. This is especially useful in the study of NEAs such as 2005 EC224 as it can help predict the occurrence of possible collisions of these with the Earth millions of years ahead of time.

Acknowledgements

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Appendices

Appendix A: Minor Planetary Center Report

COD XXX
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H5189	C2023 06 26.24626 19 02 34.16 -11 35 08.5	16.9 V	XXX
H5189	C2023 06 26.28164 19 02 34.40 -11 35 30.7	17.0 V	XXX
H5189	C2023 07 01.31899 19 03 33.81 -12 03 04.9	16.8 R	XXX
H5189	C2023 07 01.37144 19 03 33.73 -12 03 29.1	16.8 R	XXX
H5189	C2023 07 17.23299 19 05 29.96 -14 03 18.2	17.3 R	XXX
H5189	C2023 07 17.23375 19 05 29.95 -14 03 18.4	17.3 R	XXX
H5189	C2023 07 17.24354 19 05 29.93 -14 03 23.4	17.3 R	XXX

Appendix B: Cloudy Night Experiment

Introduction

Given that weather is unpredictable to some extent, it was bound for it to be cloudy on some nights, including during one of our observing shifts. Due to the non-selective scattering caused by clouds (the reason clouds appear white during the day), this makes it impossible to take usable data, especially when attempting to observe a faint asteroid. For instance, when we attempted to manually focus using one of our focus stars while a cloud was passing overhead, the image turned out extremely noisy due to all the light from the focus star being scattered across the image (see Image A). For comparison, Image B shows a focus star in the absence of a cloud, although there is some doubling (which is not due to focus).



Image A: Light scattered from a focus star through a cloud.

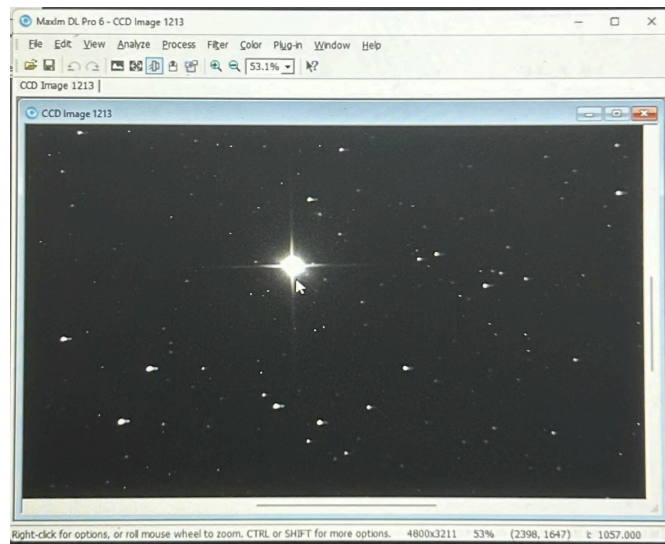


Image B: A focus star in the absence of clouds, with some streaking.

The Cloudy Night Experiments served as a way to still use the telescope to take data, only this time using dark currents (which do not require the telescope to take in light/photons from the environment) and measuring how they changed with different independent variables rather than attempting to observe the asteroid. As we had only one observing session where the sky was obscured by clouds, we were only able to complete part one of the Cloudy Night Experiments. In this experiment, we tested the effect of exposure time on the amount of dark current at a constant temperature. Our hypothesis, the so-called “Rengstorf Hypothesis,” was that the amount of dark current scales linearly with exposure time, regardless of chip type (as stated previously, the TMO 24” telescope has a CMOS chip). In theory, over time, more current will flow through the chip, and that current will show up as ‘hot pixels’ in the data.

Procedure

In order to determine the relationship between exposure time and dark current, we started with taking 1 set of 5 dark images with an exposure time of 5 seconds at a temperature of 25° C (2° below ambient temperature). Next we did another set of 5 images with an exposure time of 10 seconds at the same temperature. We continued until we had 5 sets of 5 images each, ranging in exposure time from 5 seconds to 80 seconds.

To calculate the amount of dark current in ADU in each image, we simply added up all the pixel counts for each pixel in the image. We then averaged this value over the total number of pixels to get an average pixel value for each image. Then, we averaged this average pixel value over the 5 images for each set to get an average pixel value for each exposure time.

Getting the uncertainty on each of these average dark current values was much more rigorous. First, we created noise images for each pair of consecutive images (4 total) by computing the standard deviation between corresponding pixels in these images, and mapping the result to the corresponding pixel in the noise image. After that, we took the mean of all the pixel values in each noise image, then took the mean across all 4 images to get the final noise value, which was our uncertainty.

Data

Table 3: Exposure time vs. Dark current signal

Exposure Time (sec)	Signal $\pm$ noise (ADU)
5	8557 ± 8
10	8564 ± 8
20	8571 ± 8
40	9588 ± 9
80	$10.620. \pm 10.$

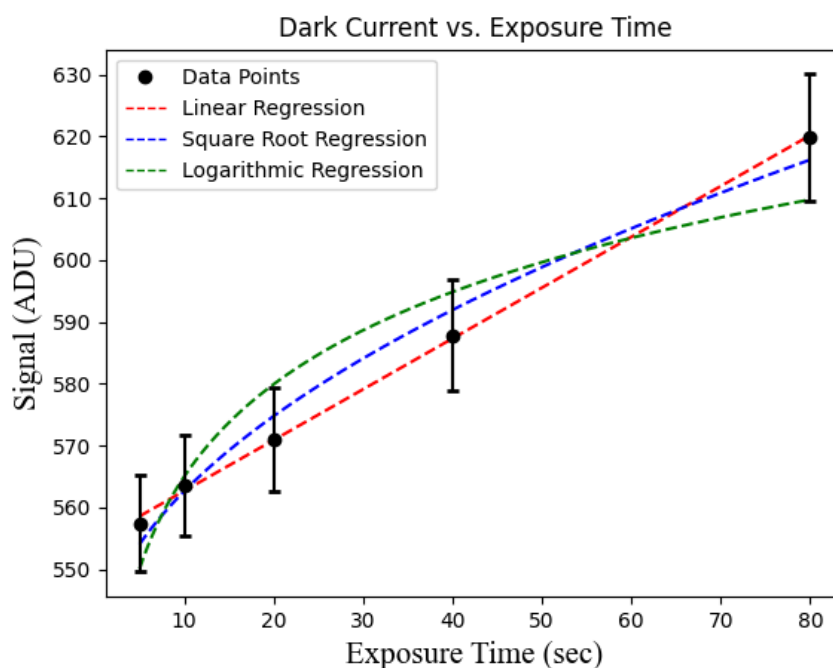


Figure 1. Dark Current vs. Exposure Time

Table 4: Regression Data

Regression	m	b	coefficient of determination
Linear ($y = mx + b$)	0.821 ± 0.015	554.4 ± 0.6	0.9989385299
Square Root ($y = m\sqrt{x} + b$)	9.3 ± 0.8	533 ± 4	0.9777312893
Logarithmic ($y = m\ln(x) + b$)	22 ± 4	515 ± 14	0.8865647732

Results and Analysis

As can be seen from the data in Figure 2. and Table 4, the linear regression best fits the plot between exposure time and dark current, with a coefficient of determination of 0.9989. Though the signal does seem to increase at a higher rate when the exposure time is lower, at least between 5 and 10 seconds, this deviation is too little for the data to match a square root or logarithmic regression. This confirms the Rengstorf Hypothesis, at least with the limited amount of data used.

Conclusion

With the given data, we fail to reject the Rengstorf Hypothesis as the data best matches a linear regression. However, we only had a limited amount of time to take data, so we were not able to take darks with as many different exposure times as we would have liked. For instance, if the slope started decreasing more, should we have had additional data points for exposure times of 160 and 320 seconds, perhaps the square root or logarithmic regressions would've better matched the data. To truly disprove the Saahil Conjecture, additional data must be collected and analyzed.

Appendix C: Orbital Element Equations

To calculate a , we used the following equation,

$$a = \left(\frac{2}{r} - \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} \right)^{-1},$$

where r is the magnitude of the position vector, and $\dot{\mathbf{r}}$ represents the velocity vector.

To calculate e , we used the following equation,

$$e = \sqrt{1 - \frac{|\mathbf{r} \times \dot{\mathbf{r}}|^2}{a^3}},$$

where $\mathbf{r}$ represents the position vector, $\dot{\mathbf{r}}$ represents the velocity vector, and a is the semi-major axis, which we just calculated.

To calculate I , we used the following equation,

$$I = \arccos \frac{(\mathbf{r} \times \dot{\mathbf{r}})_z}{|\mathbf{r} \times \dot{\mathbf{r}}|},$$

where z is the z -component of the vector h , representing the double area sweep rate. h can be calculated by crossing the position vector $\mathbf{r}$ with the velocity vector $\dot{\mathbf{r}}$.

To calculate Ω , we have the following two equations:

$$\sin \Omega = \frac{h_x}{h \sin I} \quad \cos \Omega = \frac{-h_y}{h \sin I},$$

where h_x and h_y represent the x and y components of the h vector, respectively, h represents the magnitude of the h vector, and I represents the inclination. Using these two values and the arctangent function, we can compute Ω .

To calculate ω , we have a slightly more complicated method. The angle ω represents the argument of perihelion, or the distance from the ascending node to perihelion. The angle f represents the true anomaly, or the distance from perihelion to the asteroid. We have that

$$\sin(f + w) = \frac{Z}{r \sin I},$$

where ω and f are defined above, I represents the inclination, r represents the magnitude of the position vector, and Z is:

$$Z = r \sin I \sin(f + w)$$

We also have an equation for the cosine of the sum of angles ω and f :

$$\cos(f + w) = \sec \Omega \left(\frac{X}{r} + \cos I \sin(f + w) \sin \Omega \right),$$

where ω and f are defined above, I represents the inclination, r represents the magnitude of the position vector, and X is:

$$X = r [\cos(f + w) \cos \Omega - \cos I \sin(f + w) \sin \Omega]$$

Using the equation for $\sin$ and $\cos$ of $(\omega + f)$ as well as the arctan function, we can solve for the sum of angles ω and f .

Now, we solve for angle f in order to isolate ω . We have both the $\sin$ and $\cos$ of f :

$$\cos f = \frac{1}{e} \left(\frac{a(1-e^2)}{r} - 1 \right) \quad \sin f = \frac{\mathbf{r} \cdot \dot{\mathbf{r}}}{er} \sqrt{a(1-e^2)}$$

From these equations, we can solve for angle f , then subtract that angle from $(\omega + f)$ in order to solve for ω .

Finally, we can solve for M_0 . We first solve for E , the eccentric anomaly, at the time of our middle observation, t_m :

$$\cos E = \frac{1}{e} \left(1 - \frac{r}{a} \right),$$

where e is the eccentricity, a is the semimajor axis, and r is the magnitude of the position vector. The ambiguity of E can be resolved by the fact that it is in the same half plane as f . Then, we can calculate the mean anomaly at the middle time, M_m :

$$M_m = E - e \sin E$$

Using this value, we solve for M_0 , the mean anomaly at time t_0 :

$$M_0 = M_m + n(t_0 - t_m)$$